Deep Neural Network

AIML Module 1

DNN Team

BITS Pilani WILP

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Pilani | Dubai | Goa | Hyderabad

The author of this deck, Prof. Seetha Parameswaran, is gratefully acknowledging the authors who made their course materials freely available online.

Course Logistics

What we Learn…. (Module Structure)

1. Deep learning - What, Why, where
2. Artificial Neuron and Perceptron
3. Linear Neural Network for Regression
4. Linear Neural Network for Classification
5. Deep Feedforward Neural Network
6. Convolutional Neural Network
7. Recurrent Neural Networks
8. Attention Mechanism
9. Transformer
10. Optimization of Deep models
11. Regularization for Deep models

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Books

Text book

- Dive into Deep Learning by Aston Zhang, Zack C. Lipton, Mu Li, Alex J. Smola. https://d2l.ai/chapter_introduction/index.html

Reference book

- Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville https://www.deeplearningbook.org/

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Evaluation Components and Schedule

Component	Name	Weightage	Type
EC 1	Quiz I	5 %	Pre-midsem
	Quiz II	5 %	Post-midsem
	Practice Lab Assignment I	10 %	Pre-midsem
	Situated Learning Assignment II	10 %	Post-midsem
EC 2	Mid sem Exam	30 %	Sessions 1-8
EC 3	Compre Exam	40 %	Sessions 1 -

EC1 = Quiz I + Quiz II + Assignment I + Assignment II

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Course Logistics

- Refer Taxila for the following
  - Handout
  - Schedule for Webinar
  - Schedule of Quiz, and Assignments.
  - Evaluation scheme
  - Session Slide Deck
  - Demo Lab Sheets
  - Quiz-I, Quiz-II
  - Assignment-I, Assignment-II
  - Sample QPs
- Lecture Recordings
  - Available on Microsoft teams

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Honour Code

All submissions for graded components must be the result of your original effort. It is strictly prohibited to copy and paste verbatim from any sources, whether online or from your peers. The use of unauthorized sources or materials, as well as collusion or unauthorized collaboration to gain an unfair advantage, is also strictly prohibited. Please note that we will not distinguish between the person sharing their resources and the one receiving them for plagiarism, and the consequences will apply to both parties equally.

In cases where suspicious circumstances arise, such as identical verbatim answers or a significant overlap of unreasonable similarities in a set of submissions, will be investigated, and severe punishments will be imposed on all those found guilty of plagiarism.

In case of Queries regarding the course….

Step 1: Post in the discussion forum.

- Read through the existing post and if you find any topic similar to your concern, add on to the existing discussion.
- Avoid duplication of queries or issues.

Step 2: Email to the IC at seetha.p@pilani.bits-pilani.ac.in if the query or issue is not resolved within 1 weeks time. Turn around for response to the email is 48hrs.

- In the subject pl mention the phrase ”DNN” clearly.
- Use BITS email id for correspondence. Emails from personal emails will be ignored without any reply.

PATIENCE is highly APPRECIATED :)

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What is Deep Learning?

Definitions of Deep Learning

- Deep Learning is a type of **machine learning** based on **artificial neural networks** in which multiple layers of processing are used to extract progressively higher level features from data.
- Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is **inspired by the human brain**.
- Deep learning is a machine learning technique that teaches computers to do what comes naturally to humans: **learn by example**.
- Deep learning is a subset of machine learning, which is essentially a neural network with **three or more layers**.
- Deep Learning gets its name from the fact that we add more "**Layers**" to learn from the data.

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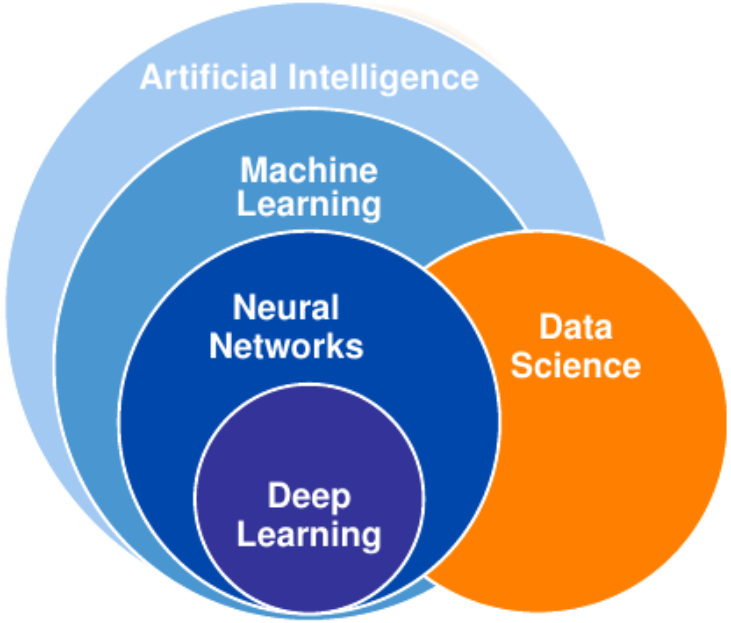
Where in AI sits DL?

AI is a general field that encompasses machine learning and deep learning, but that also includes many more approaches that don't involve any learning.



AI - ML - DL

AI : Artificial intelligence is the science of making things smart. The aim is make machines perform human tasks. Eg: Robot cleaning a room.

ML : Machine learning is an approach to AI. The machine learns or perform tasks through learning by experience.

DL : Deep Learning is a technique for implementing machine learning to recognise patterns.

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Deep (Machine) Learning

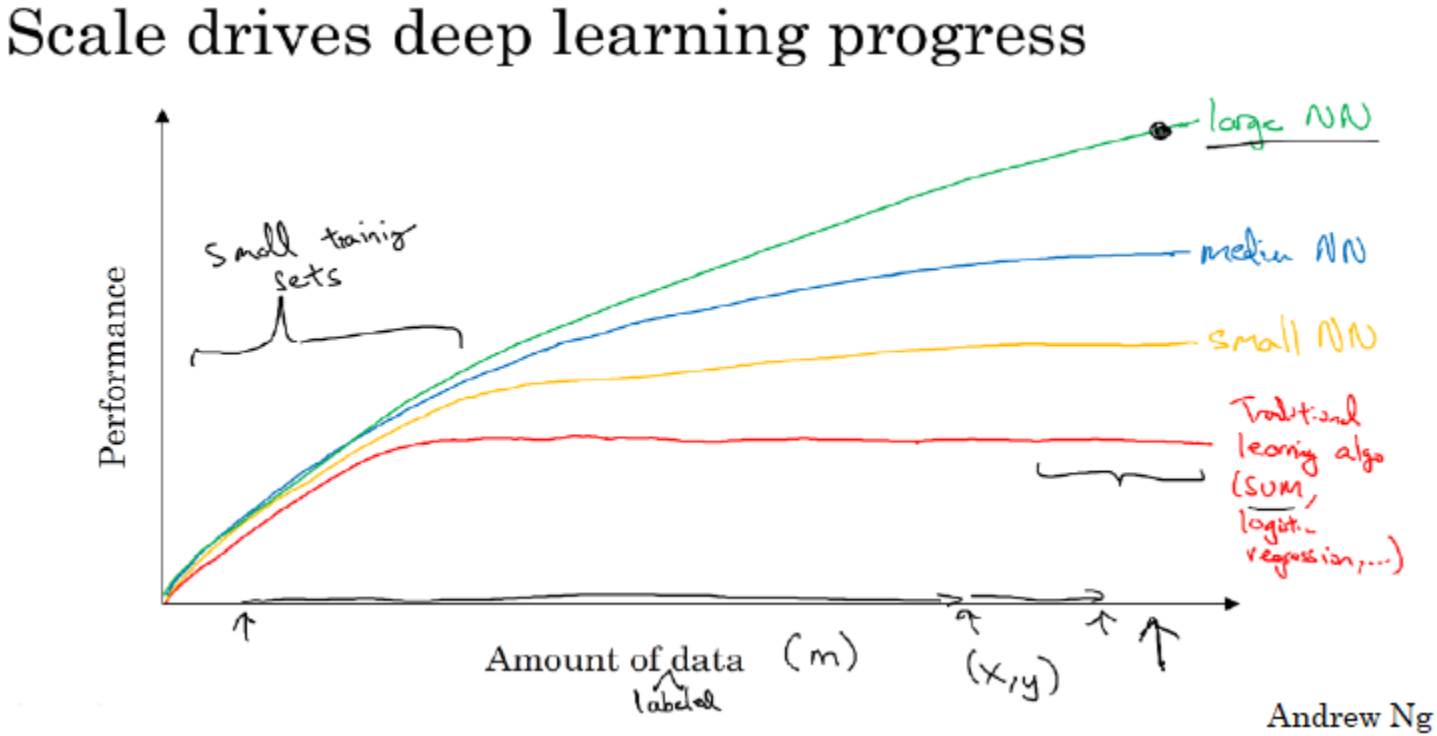
- Deep learning is a specific subfield of machine learning.
- Learning representations from data that puts an emphasis on learning successive layers of increasingly meaningful representations.
- The **deep** in deep learning stands for this idea of successive layers of representations.
- The number of layers that contribute to model the data is called the **depth** of the model.
- In deep learning, the layered representations are learned via models called **neural networks**, structured in literal layers stacked on top of each other.

Why Deep Learning?

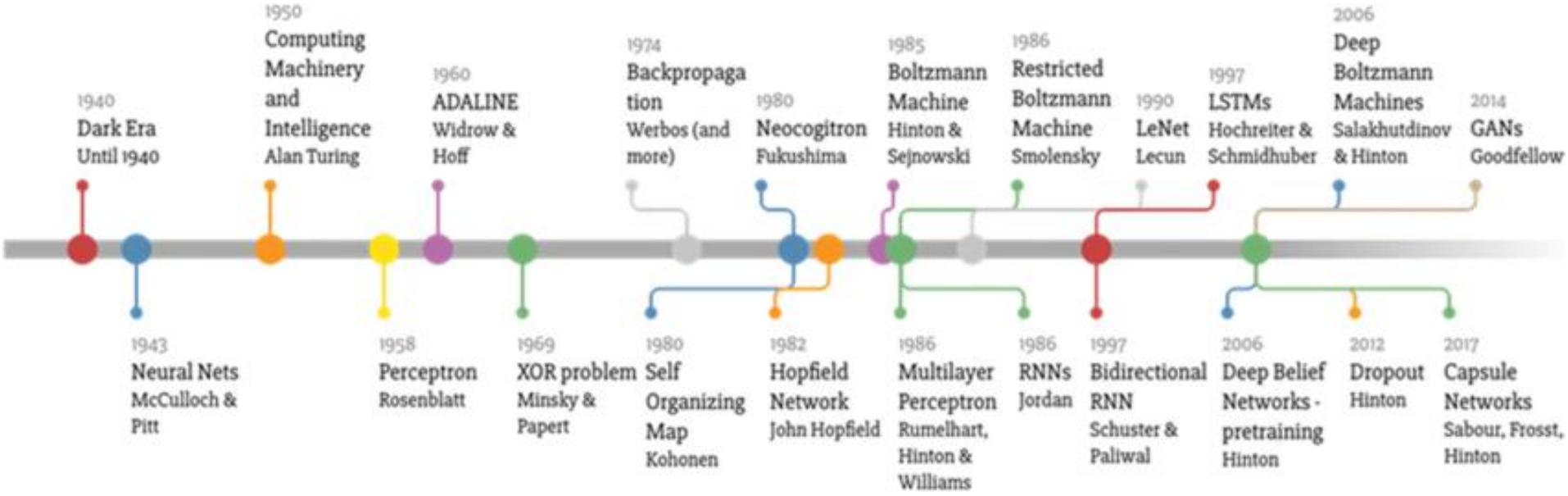
Why Deep Learning?

- Large amounts of data
- Lots and lots of unstructured data like images, text, audio, video
- Cheap, high-quality sensors
- Cheap computation - CPU, GPU, Distributed clusters
- Cheap data storage
- Learn by examples
- Automated feature generation
- Better learning capabilities
- Scalability
- Advance analytics can be applied

Why deep learning?



Deep Learning Timeline



Applications of Deep Learning

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Breakthroughs with Neural Networks

TECHNEWSWORLD EMERGING TECH SEARCH

Computing Internet IT Mobile Tech Reviews Security Technology Tech Blog Reader Service

Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

Print
Email

Image: Adobe Stock

Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.

Most Popular Newsletters News Aler

How do you feel about Black Friday and Cyber Monday?

- They're great -- I get a lot of bargains!
- The deals are too spread out -- I'd prefer just one day.
- They're a fun way to kick off the holiday season.
- I don't like the commercialization of Thanksgiving Day.
- They're crucial for the retail industry and the economy.
- The deals typically aren't that good.

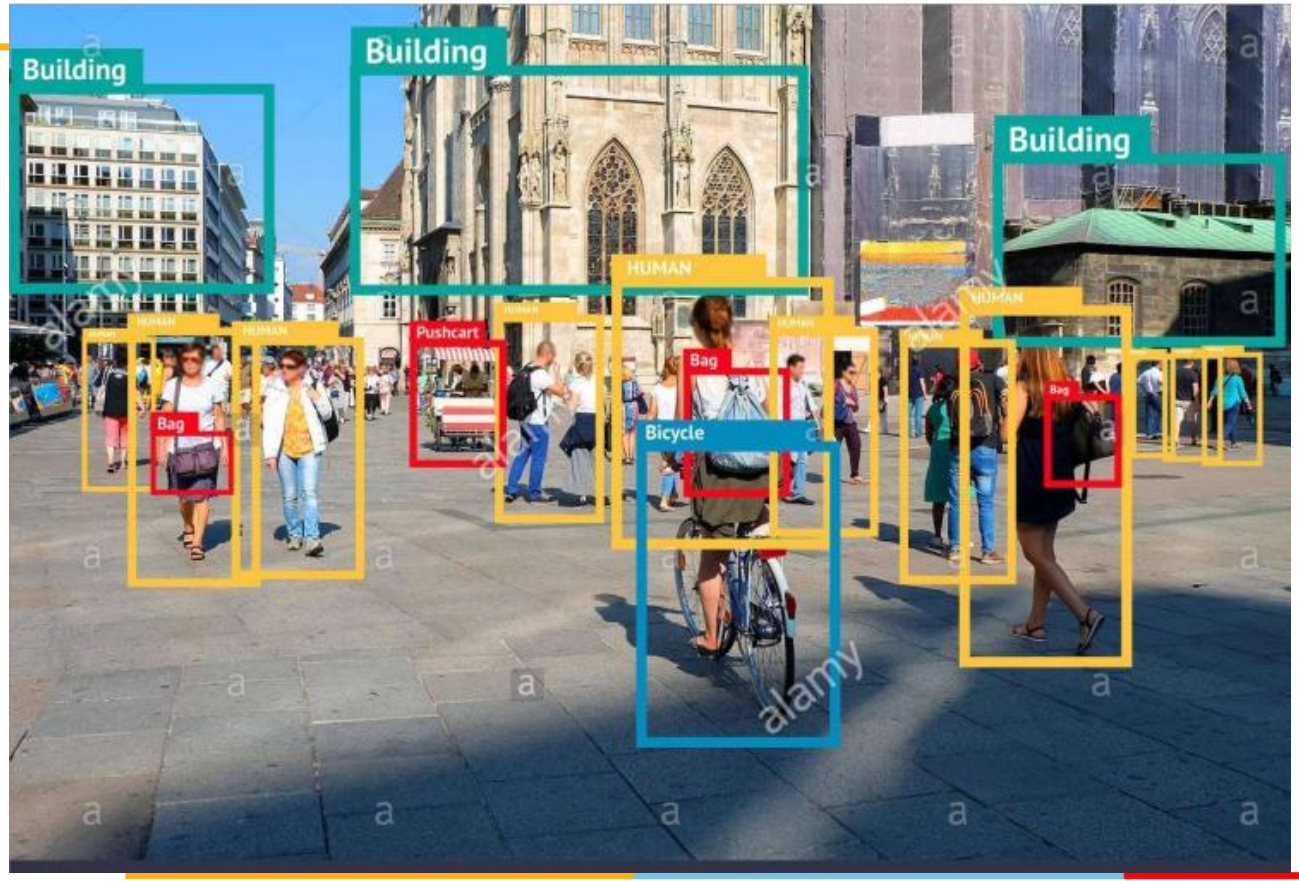
Vote to See Results

E-Commerce Times

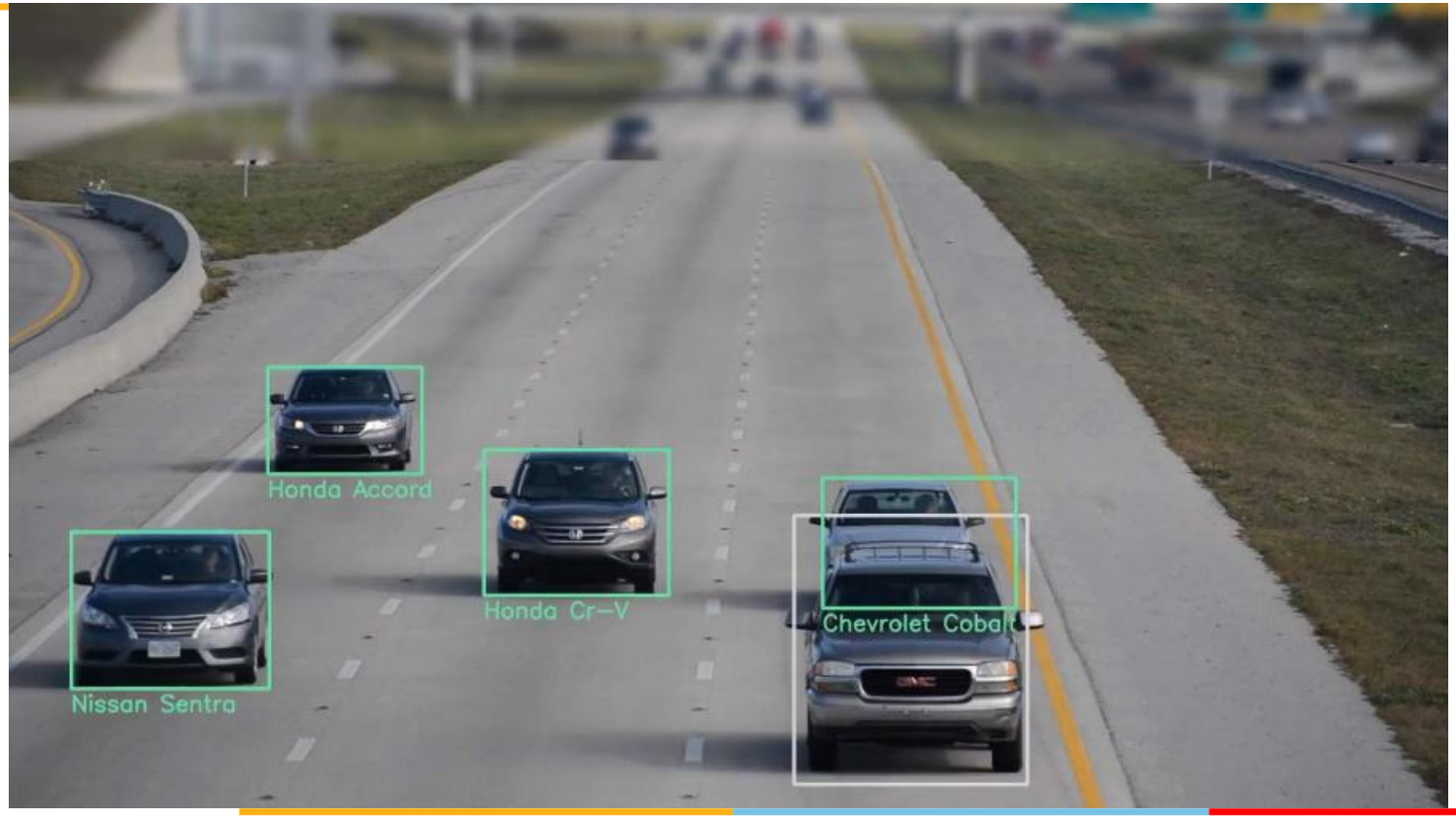
- Black Friday Shoppers Hungry for New Experiences, New Tech
- Pay TV's Newest Innovation: Giving Users Control
- Apple Celebrates Itself in $300 Coffee Table Tome
- AWS Enjoys Top Perch in IaaS, PaaS Markets
- US Comptroller Gears Up for Blockchain and

Breakthroughs with Neural Networks

Breakthroughs with Neural Networks



Breakthroughs with Neural Networks



Breakthroughs with Neural Networks

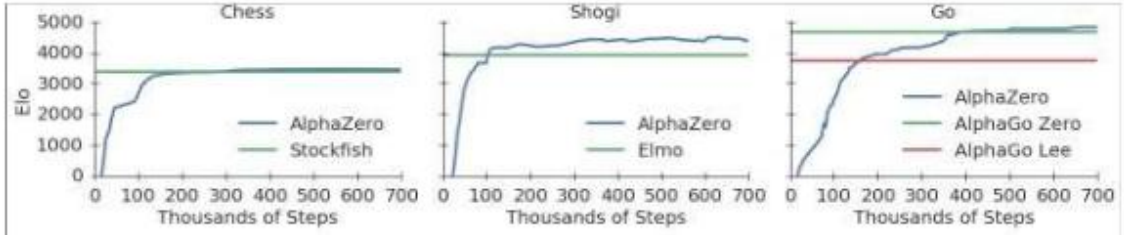
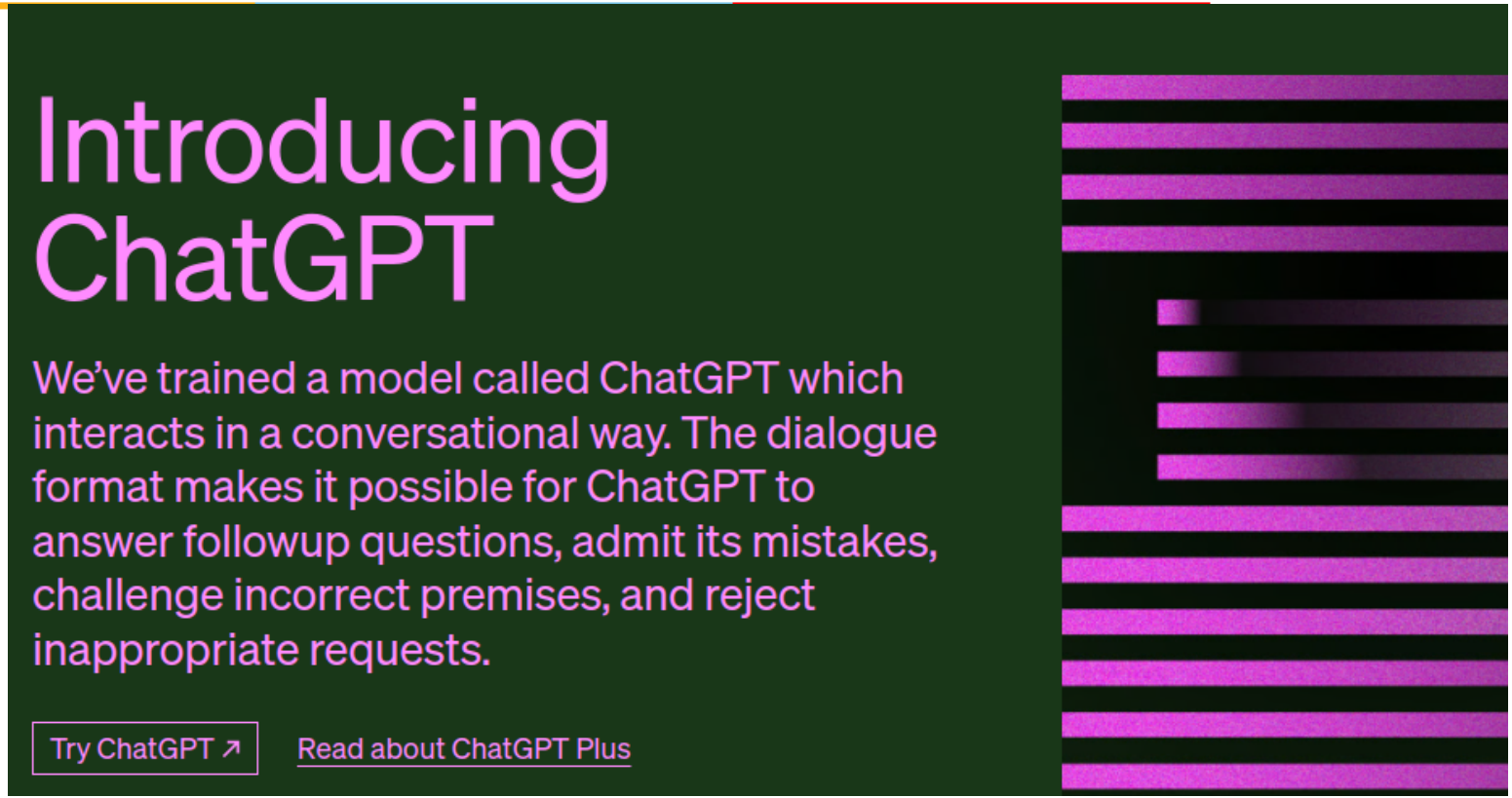


Figure 1: Training *AlphaZero* for 700,000 steps. Elo ratings were computed from evaluation games between different players when given one second per move. **a** Performance of *AlphaZero* in chess, compared to 2016 TCEC world-champion program *Stockfish*. **b** Performance of *AlphaZero* in shogi, compared to 2017 CSA world-champion program *Elmo*. **c** Performance of *AlphaZero* in Go, compared to *AlphaGo Lee* and *AlphaGo Zero* (20 block / 3 day) (*29*).

Breakthroughs with Neural Networks



Applications of Deep Learning

Application	Input	Output	Neural Network
Real Estate	House features	House Price	Std NN
Photo Tagging	Image	Text	CNN
Object detection	Image	Bounding box	CNN
Speech Recognition	Audio	Text transcript	RNN
Translation	English Text	French Text	RNN
Autonomous driving	Image, Sensors Radars	Position of other cars, objects, signals	Hybrid NN

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Many more applications….

- a program that predicts tomorrow's weather given geographic information, satellite images, and a trailing window of past weather.
- a program that takes in a question, expressed in free-form text, and answers it correctly.
- a program that given an image can identify all the people it contains, drawing outlines around each.
- a program that presents users with products that they are likely to enjoy but unlikely, in the natural course of browsing, to encounter.

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Key components of DL problem

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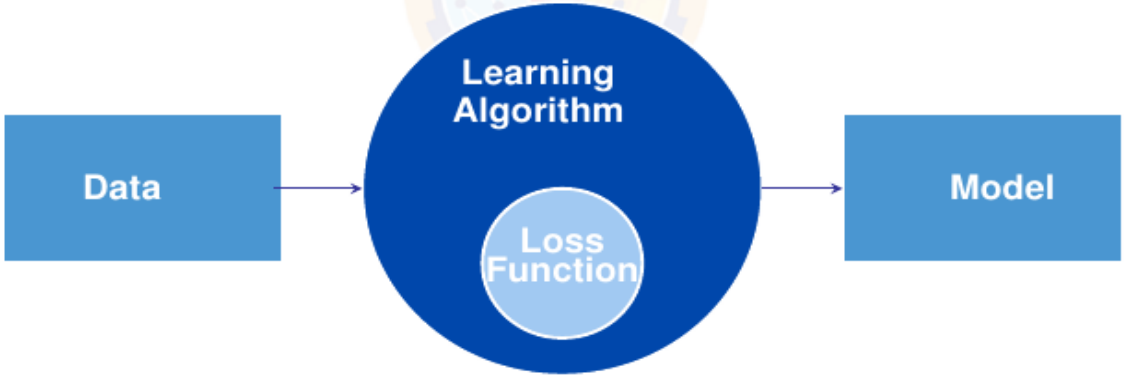
Core components of DL problem

1. The data that we can learn from.
2. A model of how to transform the data.
3. An objective function that quantifies how well (or badly) the model is doing.
4. An algorithm to adjust the model's parameters to optimize the objective function.



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1. Data

- Collection of examples.
- Data has to be converted to an useful and a suitable numerical **representation**.
- Each example (or data point, data instance, sample) typically consists of a set of attributes called **features** (or covariates), from which the model must make its predictions.
- In the supervised learning problems, the attribute to be predicted is designated as the **label** (or target).
- Mathematically, a set of *m* examples,
- We need **right** data.

$$Data = \mathcal{D} = \{X, t\}$$

1. Data

- **Dimensionality** of data
  - Each example has the same number of numerical values. This data consist of fixed-length vectors. Eg: Image
  - The constant length of the vectors as the dimensionality of the data.
  - Text data has varying-length data.

2. Model

- Model denotes the **computational machinery** for ingesting data of one type, and spitting out predictions of a possibly different type.
- Deep learning models consist of many successive transformations of the data that are chained together top to bottom, thus the name deep learning.

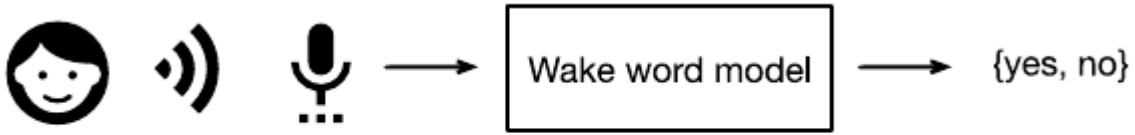

Fig. 1.1.1: Identify a wake word.

3. Objective Function

- **Learning means improving at some task over time.**
- A formal mathematical system of learning machines is defined using formal measures of how good (or bad) the models are. These formal measures are called as **objective functions**.
- By convention, objective functions are defined so that lower is better.
- Because lower is better, these functions are sometimes called **loss functions**.

3. Loss Functions

- To predict numerical values (regression), the most common loss function is squared error.
- For classification, the most common objective is to minimize error rate, i.e., the fraction of examples on which our predictions disagree with the ground truth.

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3. Loss Functions

- Loss function is defined **with respect to the model's parameters** and **depends upon the dataset**.
- We learn the best values of our model's parameters by **minimizing the loss** incurred on a set consisting of some number of examples collected for training. However, doing well on the training data does not guarantee that we will do well on unseen data. I.e **Model has to generalize better.**
- When a model performs well on the training set but fails to generalize to unseen data, we say that it is **overfitting**.

4. Optimization Algorithms

- Optimization Algorithm is an algorithm capable of **searching for the best possible parameters** for **minimizing the loss function**.
- Popular optimization algorithms for deep learning are based on an approach called **gradient descent**.

Example of the Framework

- ➔ We have to tell a computer explicitly how to map from inputs to outputs.
- ➔ We have to define the problem precisely, pinning down the exact nature of the inputs and outputs, and choosing an appropriate model family.
- ➔ Collect a huge dataset containing examples of audio and label those that do and that do not contain the wake word.

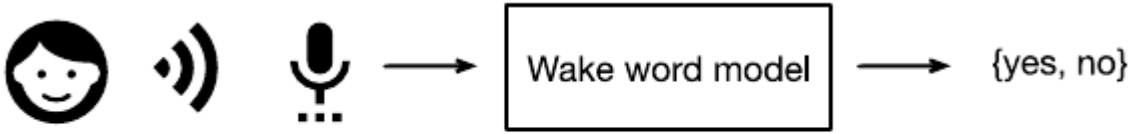

Fig. 1.1.1: Identify a wake word.

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Example of the Framework

- Create a Model
  - Define a flexible program whose behavior is determined by a number of parameters.
  - To determine the best possible set of parameters, use the data. The parameters should improve the performance of the program with respect to some measure of performance on the task of interest.
  - After fixing the parameters, we call the program a model.
    - Eg: The model receives a snippet of audio as input, and the model generates a selection among yes, no as output.
  - The set of all distinct programs (input-output mappings) that we can produce just by manipulating the parameters is called a family of models.
    - Eg: We expect that the same model family should be suitable for ”Alexa” recognition and ”Hey Siri” recognition because they seem, intuitively, to be similar tasks.

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Example of the Framework

- ➔ The meta-program that uses our dataset to choose the parameters is called a learning algorithm.
- ➔ In machine learning, the learning is the process by which we discover the right setting of the parameter coercing the desired behavior from our model.
- ➔ Train the model with d

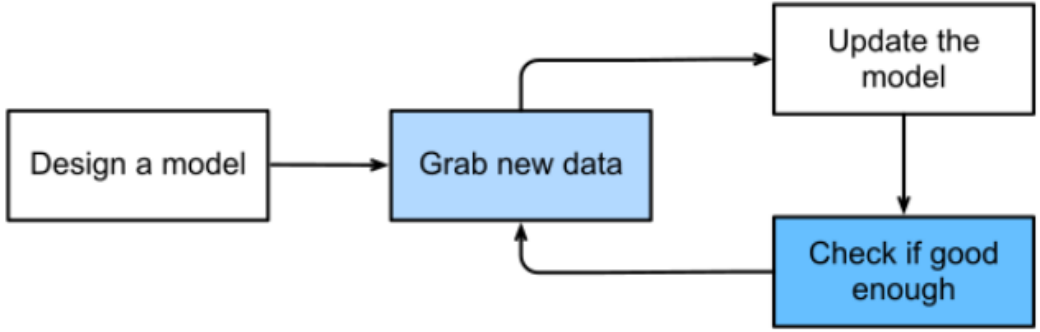

Fig. 1.1.2: A typical training process.

Kinds of Machine Learning Problems

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Learning Problems

1. Supervised Learning
   - Training by examples
   - The model is given a dataset which has the data and its labels
   - Eg: Classification, Regression
   - You will learn this in this course and Machine Learning Course
2. Unsupervised Learning
   - Situations where we feed the model a giant dataset containing only the features
   - Eg: Clustering, Generate synthetic data (uses GAN)
   - You will learn this in Advanced Deep Learning Course and Machine Learning Course
3. Reinforcement Learning
   - Develop an agent that interacts with an environment and takes actions over a series of time steps.
   - You will learn this in Deep Reinforcement Learning Course

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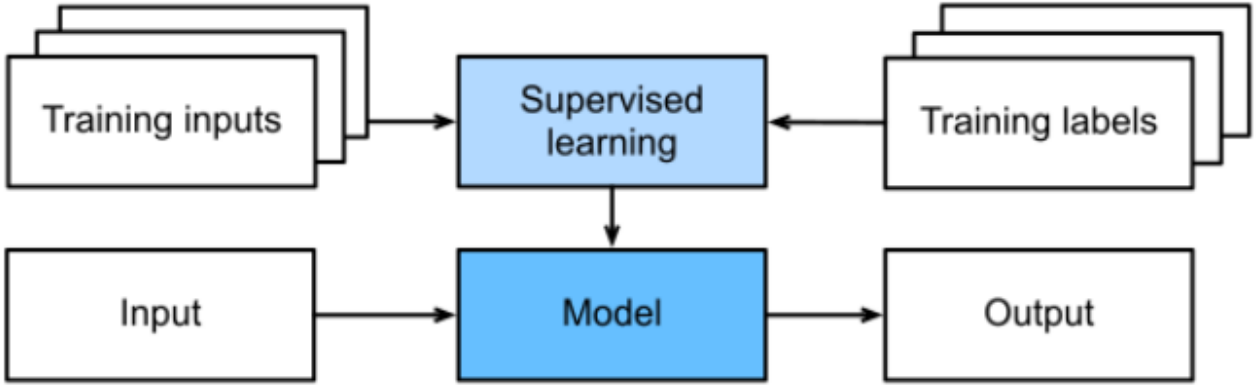
Supervised Learning

- ➔ Task of predicting labels given input features.
- ➔ Each **feature–label pair** is called an example.
- ➔ Goal is to produce **a model that maps any input to a label prediction**.
- ➔ The supervision comes into play because for choosing the parameters, we (the supervisors) provide the model with a dataset consisting of labeled examples, where each example is matched with conditional probability of a label



Supervised Learning Tasks

- Regression
  - how much or how many question
- Classification
  - Binary classification
  - Multi-class classification
  - Multi-label classification
- Recommender systems
- Sequence Learning
  - Tagging and Parsing
  - Automatic speech recognition
  - Text to speech
  - Machine translation

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Reading from TB
Dive into Deep Learning
Chapter 1

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